

AmBiDiguity does exist: UI item Direction Interpretation by Bidirectional Users

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ABSTRACT

Bidirectional user interfaces serve more than half a billion users worldwide. Despite increasing diversity-driven approaches to interface development, bidirectional interfaces still use UI items' directionality inconsistently and incorrectly. Designers should pay special attention to UI items containing ambiguous information that can be processed both from right to left and from left to right by bidirectional users. Such items are susceptible to ineffective use.

This paper reports preliminary results from a study with 1705 Arabic and Hebrew users. We investigated the directional interpretation of 16 UI items, empirically demonstrating the ambiguity problem in bidirectional interfaces and the potential influence of UI factors on how users interpret the items' directionality. While the study indicates that preventing all interpretation errors is unlikely, a large portion of those errors can be addressed by proper design.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Ambiguity, Bidirectional users, BiDi interface, Diversity, UI element direction

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1 INTRODUCTION AND BACKGROUND

Technology globalization, increasing digital accessibility, and a diversity-driven approach to interface development contributed to the emergence of interactive products in languages that differ in structure from English – the Lingua Franca of the Internet [7]. Pertinent to our research are Bidirectional (BiDi) languages that simultaneously use left-to-right (LTR) and right-to-left (RTL) directions for information representation. BiDi representation is commonly used in the languages with RTL script dominance, such as Arabic, Hebrew, Persian, Urdu, etc. [22]. LTR directionality is

used in this case to show and embed words from LTR languages (e.g., English), numbers, or notations. The total number of potential users – speakers of “BiDi languages”¹ is more than a billion² [8, 20], and at the end of August 2022, three modern BiDi languages – Arabic, Hebrew, and Persian were in the top 20 content languages on the world websites [25].

Despite the impressive presence of BiDi languages in the digital world, the state of BiDi user interfaces is far from ideal. Previous studies identified dozens of problematic UI items in BiDi interfaces and emphasized the need to study their directionality empirically with BiDi users [11, 12]. In particular, the authors call special attention to investigating a class of UI items that we term “*amBiDiguous*.” *The common characteristic of amBiDiguous items is that without any previous knowledge or explicit contextual information, BiDi users are uncertain if they should process the items from right to left or from left to right* (see examples of such items in Fig.1). Unfortunately, the use of UI items with ambiguous information is common in popular BiDi interfaces [12].

Although HCI practice deals with the positive impact of ambiguity in interface design [5], there are many cases when ambiguity leads to frustration and user errors [10]. The BiDi user interface is an example of such undesirable consequences. An amBiDiguous information representation reduces BiDi users' ability to understand correctly a message conveyed by the designer. Despite the importance of the topic, the world's leading web companies, Apple, Google, and Microsoft, do not address the ambiguity information prevention issue in the bidirectional guidelines [2, 13-15].

Moreover, to the best of our knowledge, with one exception, there is no research on ambiguous information interpretation in BiDi interfaces and the potential effect of UI factors on them. The exception is a study of UI item direction preference [12]. However, that study investigated the directional interpretation of only four ambiguous UI items, using a limited number of participants in a homogeneous sample of Hebrew speakers.

The current paper aims to fill the gap in amBiDiguous UI item investigation. We asked the following research questions: (1) *What is the interpreted directionality for common “amBiDiguous” UI items in BiDi interfaces?* (2) *What is the potential influence of UI factors' manipulation on UI item direction interpretation?* Our analyses thus far provide empirical evidence for the amBiDiguity problem existence. Future analyses (outside the scope of this paper) will analyze other factors that influence the interpretation of amBiDiguous UI items. We believe it is important to share these early findings with

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¹Here and below, “Languages that mix RTL and LTR script in text representation.”

²The number does not include speakers of languages with LTR or top-to-bottom writing systems that use RTL strings or RTL reading order for historical reasons and sometimes to simulate ancient writing, for example, Chinese and Japanese [6, 21].



Figure 1: Examples of amBiDiguious UI items in BiDi interfaces. (a) Next and Previous buttons in a student exams' platform checking. The Hebrew label reads, "Search for a user." Users might interpret the right and left arrows either as "next" or "previous." (b) Promotional text on a drugstore website indicating that the user will receive a discount if she purchases a certain number of items. Read RTL (according to Hebrew text direction), it is interpreted as "buy one, get two"; Read LTR, it is interpreted as "buy two, get one." The label reads, "Valid until 14/12/2022." (c) Year range input (after entering "2015") in an Arabic version of the Google scholar website (the information is presented in a similar way in other BiDi versions). The label is "Custom range." Users can interpret the information as "until 2015" or "from 2015."

the HCI community to raise awareness of the amBiDiguity problem and prevent interpretation errors by BiDi users.

2 METHOD

We conducted an online experiment with Arabic and Hebrew speakers. The research was approved by the university's IRB and was carried out during 2022. The study was conducted in cooperation with the National Digital Israel Initiative³. Due to complex manipulations within the study, the experimental application was developed by a commercial firm according to our specifications and under our supervision. Screenshots of all items and stimuli from both samples are provided in the Supplementary materials.

2.1 Procedure

The study was conducted in Arabic or Hebrew (depending on the sample subgroup). The participants were recruited through social networks and a participant pool of the firm administering the experiment. Participation was voluntary; participants were compensated with the equivalence of about \$3 following the completion of the experiment. Participants were allowed to participate only once.

The participants received a link to the application. After clicking on it, the application presented an explanation of the study goals and terms. The participants completed the consent form before being allowed to continue the study. Participants worked at their own pace, and there were no time constraints. To register the participants' intuitive reactions, we did not provide an opportunity to revise previous answers. All items were presented one by one.

2.2 Study design

The study consisted of three parts: (1). *Introduction*: descriptions of the experiment goals, the collected data, and the consent form; (2). *Demographics* - comprised of items collecting background information (e.g., age, native language, dominant hand, occupation, et cetera). The data obtained in this part could be used to analyze the influence of individual factors; (3). *Experiment* - included 32 experimental tasks and one attention task presented randomly, each on a single, separate screen. The design of all tasks and UI items in the Experimental part was based on real-life examples we found in popular BiDi interfaces (examples shown in Fig. 1); or on reports of users and HCI professionals (e.g., in the Hebrew-speaking Facebook group "Technologically challenged people that are harming businesses rating" [26]). *Sixteen of the 32 tasks investigated amBiDiguious items; in the current paper, we focus only on their analysis* (see amBiDiguious UI items list in Appendix 1). We provide a list of focal items and used tasks in the Supplementary materials for study reproducibility.

In developing the experiment, we considered that UI items could be used for two purposes - to convey information and to perform an action. Therefore, we developed two task types: (1). *Tasks for information understanding*. For example, the user saw a tandem of "Undo and Redo" buttons and was asked what action would be performed if clicking on one of them (see Appendix 2). Tasks of this type were shown with two answers differing in UI item direction interpretation. (2). *Tasks for action performing*. For example, the participants saw a pair of text fields and had to enter a credit card's expiration date in an ll/rr format (see Appendix 3). The user's interpretation of the directionality of the format was inferred by whether ll/rr were mapped to mm/yy or yy/mm. The tasks were

³A government initiative to improve citizens' accessibility to government services and to information and communication technologies.

presented in the same format in mobile and desktop versions and included three or four elements, depending on the task. Each task included the following elements, from top to bottom:

- Explanation of the task.
- The UI item⁴ - was shown on a white background and had a text label. We selected items based on a list of challenging UI items in BiDi interfaces [12] and recommendations for UI items' prioritization in the BiDi research [11]. The items belonged to four different logical groups (and six sub-groups) [12], so the influence of these factors could be further analyzed.
- Two answer options (in "Information understanding" tasks' type).
- A "Next" button. The button had an unavailable status until the participant responded to the task.

All task elements were always visible on the screen, with the exception of the "Information understanding" tasks, in which the "Next" button appeared after the answer selection. An example of a task with four elements is shown in Appendix 4.

To test the UI design factors' influence on UI items direction interpretation, we developed *four conditions* differing in the text alignment and the UI item direction. We chose these factors because they are the most considered UI factors in the existing BiDi guidelines [1-4, 13-15]. "Full right" and "Full left" conditions simulated a text alignment that agreed with the UI items' direction. "Full Right" simulated a complete RTL interface (RTL alignment, RTL UI item direction), whereas "Full Left" simulated a complete LTR interface. The Center and Random conditions simulated some inconsistency between the text alignment and the UI item directionality (see Appendix 5 for more information). For ambiguous UI items, context manipulation was on text and UI item alignment. The answer options and the "next" button were centered on all conditions. Each participant was assigned to one of four conditions.

The data was analyzed by Python Statsmodels [18] package and the SPSS Statistics platform, version 28.0.1. In each task we recorded the participants' responses and then coded their direction (LTR or RTL).

2.3 Participants

The study was completed by 1820 participants who correctly answered the attention check item. We conducted a data cleansing procedure, following which we discarded 115 records for various QA or missing data issues. The remaining 1705 bidirectional participants' average age was 40.27 years (SD = 15.610; range = 18 to 82). The average age of 620 participants in the Arabic sample was 33.94 years (SD = 11.26; range = 18 to 77). The average age of 1085 participants in the Hebrew sample was 43.88 years (SD = 16.570; range = 18 to 82). The difference in average age between the groups is commensurate with the average age difference in technology usage by Arabic and Hebrew Israelis [9]. The proportion of women was 52.7%. The participants varied in individual characteristics (e.g., in LTR language knowledge, occupation, dominant hand, et cetera). Appendix 6 summarizes the participants' demographic data.

⁴In the attention task, we used an image with two tomatoes instead.

3 RESULTS AND DISCUSSION

3.1 Interpreted direction of amBiDiguous UI items

BiDi users might face uncertainty when attempting to infer the right direction in which they should process amBiDiguous items. Here, we estimate this uncertainty for each item as the proportion of participants who interpreted direction as RTL vs. LTR. Any deviation from 100% (for either LTR or RTL interpretation) reflects uncertainty and interpretation error probability; 50% interpretation processing reflecting complete uncertainty. The proportion of participants who interpreted the item as requiring processing from RTL can be considered an indicator of the sample's preference that the item should be designed in the RTL direction and vice versa. Finally, this data could be used as a basis for design guidelines regarding the directionality of specific amBiDiguous items.

To test the statistical significance of the interpreted UI items' directions, we performed a Z-test for a single proportion (50% is a target value), with a confidence level of 99.8% - calculated with Bonferroni correction. Due to the potential influence of linguistic factors on UI item direction interpretation [24] and differences in interface usage between Arabic and Hebrew-speaking Israelis [9], this analysis was done separately for each language (Arabic, Hebrew).

We observed a tendency ($p > 0.001$) to interpret an LTR direction for 11 UI items in the Arabic sample and 10 UI items in the Hebrew sample. The LTR interpretation was observed mainly in UI items that include *numerical and date strings* (see Table 1 and Fig. 2). A potential explanation for the LTR interpretation may lie in compatibility with previously learned perceptual concepts, such as that numerical strings should be read from LTR as in mathematics. Similarly, Gregorian dates are written LTR in BiDi languages. We assume that the preference of "numerical" UI items might change to RTL when shown unambiguously (e.g., with RTL labels "from" and "to" for ranges). This issue requires further study.

We found a tendency for RTL interpretation for three UI items in the Arabic sample and four in the Hebrew sample ($p > 0.001$). These UI items (*event flow image*, *event flow text*, *process widget*, *next & previous pair*) include only images and RTL labels. Users were split at almost "50-50" level regarding the directional interpretation of the "undo and redo" pair in both samples, the "backward and forward" pair in the Hebrew sample, and the "year range string" in the Arabic sample.

It should be noted that the participants from both samples show the same directional interpretation for all UI items except for the "Event flow image" item (see the task in Appendix 7). Arabic-speaking participants tended to interpret the directionality from LTR as "the girl brushed her teeth and then went to sleep" (65 % LTR). In contrast, 61 % of the Hebrew-speaking participants interpreted it in the RTL direction, as "the girl slept and then brushed her teeth." However, this difference could be due to the unique cultural characteristics of this specific image (i.e., the teeth brushing" sequence) and may not be generalized to other sequences. Clarifying this issue would require future research to test other instances of the "Event flow image" item.

Two important observations emerge from this analysis: First, even when the UI item directional preference was clear, there was

Table 1: Direction interpretation of amBiDigous UI items (all significant differences are at $p < 0.001$)

#	UI item	Sample	General LTR Interpretation ^a	LTR interpretation in Conditions				Difference between conditions ^b
				Full right	Full left	Center	Random	
1	“Backward & forward” buttons (I)	A	56%, $z=3.052$	53%	66%	53%	53%	n/s
		H	53%, $z=2.034$, n/s	43% ↔	59%	57%	55%	16.75
2	Date range input (A)	A	80%, $z=14.409$	71%	88%	81%	80%	n/s
		H	88%, $z=24.805$	83%	90%	91%	88%	n/s
3	Date range string (I)	A	81%, $z=15.502$	81%	86%	80%	78%	n/s
		H	88%, $z=25.088$	86%	90%	89%	88%	n/s
4	Event flow image (I)*	A	65%, $z=7.309$	52%	76%	63%	68%	19.65
		H	39%, $z=7.377$	23% ↔	56%	36%	42%	66.42
5	Event flow text (I)	A	32%, $z=8.802$	15%	44%	37%	35%	33.16
		H	30%, $z=13.206$	15%	46%	28%	32%	68.71
6	Expiration date input (A)	A	89%, $z=18.328$	84%	91%	91%	90%	n/s
		H	98%, $z=31.089$	96%	98%	98%	98%	n/s
7	Hour range input (A)	A	77%, $z=13.302$	66%	87%	83%	74%	21.80
		H	87%, $z=24.141$	83%	90%	87%	88%	n/s
8	Hour range string (I)	A	95%, $z=22.169$	93%	95%	96%	95%	n/s
		H	93%, $z=28.108$	92%	93%	94%	92%	n/s
9	“Next & previous” buttons (I)	A	26%, $z=12.098$	13%	32%	25%	33%	21.32
		H	24%, $z=16.910$	14%	32%	27%	26%	24.93
10	Process widget (A)	A	37%, $z=6.014$	22% ↔	53%	39%	36%	27.20
		H	27%, $z=15.225$	16%	41%	21%	30%	50.77
11	Sale 1+2 (I)	A	57%, $z=3.701$	45% ↔	65%	67%	55%	19.71
		H	60%, $z=6.290$	50%	67%	60%	61%	17.37
12	Sale 1+3 (I)	A	56%, $z=3.133$	48%	61%	62%	55%	n/s
		H	57%, $z=4.556$	50%	65%	56%	57%	n/s
13	Sale 2+3 (I)	A	61%, $z=5.462$	51%	64%	67%	62%	n/s
		H	64%, $z=9.199$	54%	70%	66%	68%	19.08
14	“Undo & redo” buttons (I)	A	49%, $z=0.562$, n/s	38% ↔	56%	48%	55%	n/s
		H	53%, $z=1.791$, $p=0.037$	46% ↔	61%	53%	52%	n/s
15	Year range input (A)	A	75%, $z=12.228$	62%	86%	74%	77%	23.03
		H	83%, $z=21.422$	76%	88%	83%	84%	14.63
16	Year range string (I)	A	54%, $z=2.088$, n/s	45% ↔	68%	54%	50%	18.19
		H	59%, $z=5.616$	57% ↔	69%	48%	60%	26.37

Legend: (I) Informational task type. (A). Actionable task type. (*) Opposite direction interpretation in samples. (a) Z-test for a single proportion (LTR interpretation > 50%) with a confidence level of 99.8%. (b) Kruskal-Wallis H (df=3). (↔) Opposite directional interpretation between Full Right and Full Left conditions.

still considerable room for the opposite interpretation. Out of the 32 comparisons (16 in Arabic and 16 in Hebrew), only in eight was the proportion of preference for one processing direction greater than 80% (see Table 1). Thus, it is impossible to speak about a unanimous directional interpretation of the study’s items. Instead, research should shift into asking how to design mechanisms for reducing amBiDiguity. Second, while there were considerable differences in directional interpretations between items, items belonging to the same group or sub-group were interpreted similarly. This finding indicates a potential influence of the nature of a UI item and the group to which it belongs on its directional interpretation. We will present an analysis of the data related to this issue in a subsequent publication.

3.2 Influence of UI item and text alignment

We tested whether interpretations of UI items differed between experimental conditions in the Arabic and Hebrew-speaking samples separately. For this purpose, we performed a set of Kruskal-Wallis H tests (experimental condition as an independent variable, interpretation of UI item direction as the dependent variable, confidence level 99.8% - calculated with Bonferroni correction). The tests found significant differences in direction interpretation between conditions in 17 out of 32 comparisons in both samples (see Table 1 and Figure 3). With very few exceptions, the “Full right” condition strengthened the RTL interpretation and demonstrated the highest level of RTL interpretation for amBiDigous items in both Arabic and Hebrew samples. The “Full left” condition had the opposite effect in both samples.

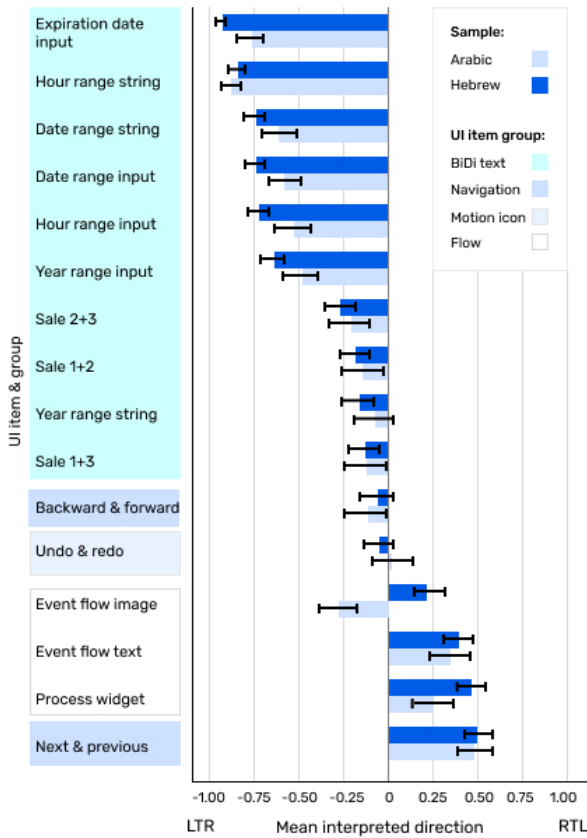


Figure 2: Mean ambiguous UI items' interpreted direction.

We observed that manipulations of the context alignment could sometimes reverse the direction interpretations. For example, in the “Full left” condition, the proportion of LTR interpretation for “Event flow image” in the Hebrew sample was 56%. In contrast, it was RTL in other conditions, with maximal RTL tendency in the “Full right” condition (23% interpreted as LTR).

The results stress the importance of correct BiDi design. Determining the optimal text alignment and UI item direction is indeed a responsibility of HCI professionals. The UI context can affect directionality interpretation, and thus it must be considered carefully in BiDi design. We also suggest to consider investigating other context-creating design factors, e.g., label boldness or color manipulation. Studying these factors can uncover signifiers to disambiguate UI item directionality and contribute to BiDi design guidelines.

4 RECOMMENDATIONS AND OPEN QUESTIONS

Our findings demonstrate the amBiDiguity problem. To reduce amBiDiguity in BiDi interfaces, we strongly recommend providing signifiers regarding the correct directionality. By signifiers, we refer to Norman's notion of “any perceivable indicator that communicates appropriate behavior to a person” [17, p. 14]. We suggest the

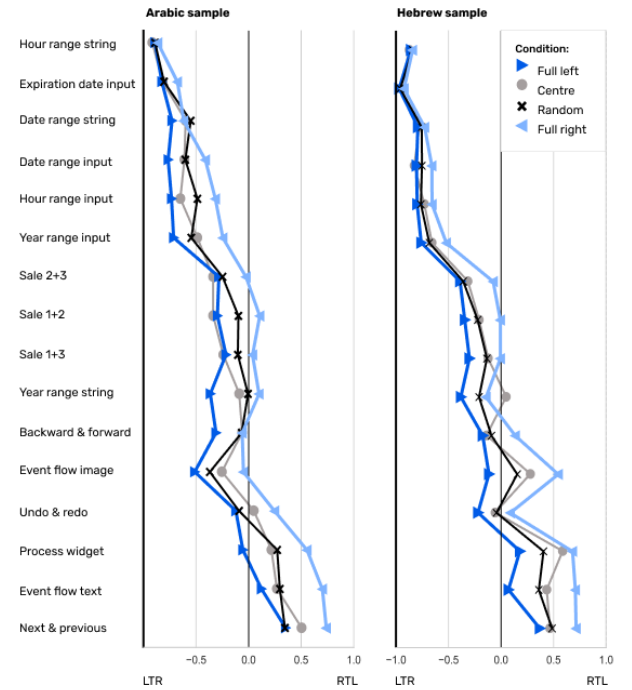


Figure 3: Mean UI item direction interpretation in study conditions

following measures for amBiDiguity reduction. (1) *Provide a label*. Although using labels is a self-evident, trivial, and imperative solution for accessible UI design [23], they are absent from many BiDi interfaces. We believe that providing labels is a mandatory requirement for reducing amBiDiguity. (2) *Show in-context descriptions or hints* for essential information. While a description should always be visible [19], it is particularly valuable for certain BiDi items. An example of using hints to explain information direction could be explicitly writing (in the BiDi language) “Buy 2 get 1 free” for the sales promotion label “2+1” (see fig.1b). The label “2+1” is unmistakable in an LTR language, but is ambiguous in BiDi language. (3) *Use visual cues*, such as icons, animation, color hierarchy, etc., to describe the UI item directionality. This design solution is especially relevant for “design-dependent” UI items such as the event flow text [11, p.4]. For example, adding directional arrows would remove directional ambiguity in this item. (4) *Stress the correct direction by content alignment*. To highlight RTL UI item direction, we recommend aligning the content to the right; for LTR UI item direction, consider centering content alignment as another appropriate text alignment in BiDi interfaces.

In fig. 1(c) we showed a frequently used yet ambiguous presentation of the year-range input item. This item is reproduced in fig. 4(a) alongside examples for addressing this problem.

We also note that the optimal solution for reducing amBiDiguity may depend on the particular platform. For example, placing labels above the field is most suitable for mobile platforms [16]), and should consider findings from quantitative and qualitative research with BiDi users.



Figure 4: A variant of design solutions to reduce amBiDiguity, using the example of year input. (a) Ambiguous information representation, identical to figure 1(c); (b) Adding the labels “From year” and “To year” at the top of the field, suitable for mobile platforms; (c) Adding the labels “from” and “to” on the side of the field.

Acknowledging the need to use signifiers to reduce ambiguity opens a question – “Which type of visual cues is the most important in the BiDi interface?” This question is especially relevant for the minimalist design trend that discards “unnecessary” elements. Are the critical elements in the BiDi interface just as important in the LTR design, or do they need to be redefined explicitly in BiDi design guidelines? These questions highlight the importance of further amBiDiguity investigation with a control group of LTR speakers. We also recommend using qualitative methods, which can broaden the understanding of the results obtained in the current study and suggest strategies for making decisions about ambiguous information. We call on the HCI community to join us in this direction.

5 CONCLUSION

We use the term “amBiDiguous” UI items to denote a common situation in BiDi interfaces in which users are uncertain whether to process an item from LTR or from RTL. This online experiment with Arabic and Hebrew speakers provides empirical evidence for the ambiguity of 16 amBiDiguous items. Users face varying degrees of uncertainty when interpreting the directionality of such items, which might lead to user error and frustration. The BiDi UI developer’s responsibility is to reduce potential amBiDiguity. One of the ways to do this is to provide signifiers regarding the correct directionality. The current work describes partial results of the amBiDiguity study. To the best of our knowledge, this is the first study that aimed to empirically measure the directional ambiguity of BiDi UIs on a large sample of Arabic and Hebrew users. Additional research is needed to replicate and validate our findings and to explore additional aspects of the topic. For that, we share our findings and offer the study’s method as a basis for future investigations.

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APPENDICES

A1 AMBIGUOUS UI ITEMS PRESENTED IN THE STUDY

Table 2: AmBiDigous UI items presented in the study.

UI item	Task example (Centered condition)		
	Arabic sample	Hebrew sample	Task explanation translation
Group “BiDirectional text” Sub-group “Ranges”			
Date range (A, I)	يجب تحديد مجال التاريخ لامرأة خرجت لإجازة مرض من تاريخ 3 وحتى 8 آذار في أي تاريخ خرجت إلى إجازة مرض في شهر آذار؟ 00 - 00	יש להגדיר את טווח התאריכים עבור אישה שלקחה ימי מחלה מה-3 ועד ה-8 במרץ באילו תאריכים לקחת ימי מחלה במרץ? 00 - 00	“Set the date range as a woman who took sick leave from March 3rd to 8th.” The label in bold reads: “On what dates do you take sick days in March?” LTR interpretation: a string 03-08. RTL interpretation: a string 08-03.
	بحسب الإعلان، هل سيكون الموقع مغلقاً في شهر تموز؟ الموقع سيكون مغلقاً بهدف الصيانة بين التواريخ: 2.2-8.8	לפי המודעה, האם בחודש יולי האתר יהיה סגור? האתר יהיה סגור לשיפונים בין התאריכים: 2.2-8.8	According to the announcement – will the site be closed in July? The label in bold reads: “The site will be closed for renovations between the dates: 2.2-8.8” LTR interpretation: “yes;” RTL interpretation: “no.”
	يجب تحديد مجال الساعات لشخص عمل من الساعة 7:00 وحتى الساعة 17:00 في أي ساعات عملت اليوم؟ 00 - 00	יש להגדיר את טווח השעות עבור אדם שעבד בין 7:00 ועד 17:00 באיזה שעות עבדת היום? 00 - 00	Set the hour range as a person who worked from 07:00 to 17:00. The label in bold reads: “What time did you work today?” LTR interpretation: “07-17”; RTL interpretation: “17-07”.
Hour Range (A, I)	بحسب الإعلان، ما هو المجال الزمني الذي يكون فيه الوقوف بدفع؟ الوقوف بدفع بين الساعات: 09:00-21:00	לפי המודעה, מהו טווח הזמנים בו החניה בתשלום? החניה בתשלום בין השעות: 09:00-21:00	What is the time for paid parking according to the announcement? The label in bold reads: “Parking is paid between the hours: 09:00-21:00” LTR interpretation: “from nine AM till nine PM”; RTL interpretation: “from nine PM till nine AM”.
	يجب إدخال سنوات العمل لشخص عمل من العام 2015 وحتى العام 2018 في أي سنوات عملت في مكان العمل؟ 0000 - 0000	יש להזין את שנות העבודה עבור אדם שעובד מ-2015 ועד 2018 באילו שנים עבדת במקום העבודה? 0000 - 0000	“Enter work years for a person who worked from 2015 to 2018. The label in bold reads, “In which years did you work at the workplace?”. LTR interpretation: a string 2015-2018; RTL interpretation: a string 2018-2015.
Year range (A, I)			

Table 2: continued

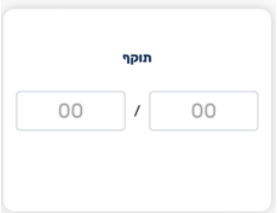
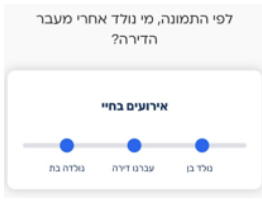
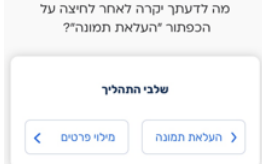
UI item	Task example (Centered condition)		
	Arabic sample	Hebrew sample	Task explanation translation
Expiration date input (A)	في الصورة العرضة أمامك، يمكن البحث عن سيارات بحسب سنة الإنتاج. براك، أي سيارات سيتم عرضها بعد الضغط على زر "بحث"؟ 	בתמונה המוצגת ניתן לסנן את תוצאות החיפוש לפי שנת ייצור. איילו רכבים יוצגו אחרי הלחיצה על הכפתור "חיפוש"? 	In the displayed image, you can filter the search results according to the manufacturing year. Which cars will be displayed after clicking the "Search" button? The label reads: "manufacturing year", the button label reads "Search" LTR interpretation: "before 2016"; RTL interpretation: "after 2016".
	Sub-group "Gregorian Date string"		
Expiration date input (A)	يجب اختيار شهر كانون الثاني 2023 الصلاحية 	יש להזין את חודש ינואר 2023 תוקף 	"You should enter "January 2023". The label in bold reads, "Expiration". LTR interpretation: entered MM/YY. RTL interpretation: entered YY/MM.
Sale 1+2 (I)	Sub-group "Containing the offer "Sale "X+Y"		
	بحسب الإعلان، على كم قطعة نحصل كهدية؟ حملة 1+2	לפי המודעה, כמה פריטים מקבלים במתנה? מבצע 1+2	"According to the advertisement, how many items will you get as a gift?" The label in bold reads, "Sale". LTR interpretation: "two items". RTL interpretation: "one item".
Sale 1+3 (I)	Sub-group "Containing the offer "Sale "X+Y"		
	بحسب الإعلان، على كم قطعة نحصل كهدية؟ حملة 1+3	לפי המודעה, כמה פריטים מקבלים במתנה? מבצע 1+3	"According to the advertisement, how many items will you get as a gift?" The label in bold reads "Sale". LTR interpretation: "three items". RTL interpretation: "one item"
Sale 2+3 (I)	Sub-group "Containing the offer "Sale "X+Y"		
	بحسب الإعلان، على كم قطعة نحصل كهدية؟ حملة 2+3	לפי המודעה, כמה פריטים מקבלים במתנה? מבצע 2+3	"According to the advertisement, how many items will you get as a gift?" The label in bold reads "Sale". LTR interpretation: "three items". RTL interpretation: "two items"
Event flow image (I)	Group and sub-group "Time flow"		
	بحسب الرسمة، ما هو ترتيب العمليات الظاهر في الصورة؟ صور نمط حياة سمر 	לפי האיור, מה סדר הפעולות המתואר בתמונות? תמונות מחייה של טל 	"Based on the illustration, what is the order of actions described by the images?" The label in bold reads "Images from Tal's life." LTR interpretation: "brushed her teeth first". RTL interpretation: "slept first"

Table 2: continued

UI item	Task example (Centered condition)		
	Arabic sample	Hebrew sample	Task explanation translation
Event flow-text (I)			“According to the image, who was born after the moving to a new apartment?” The label in bold reads, “Events of my life.” The labels on the timeline (from LTR: a daughter was born; moving to a new apartment; a son was born). LTR interpretation: “son”. RTL interpretation: “daughter”.
Process widget (A)			“You should mark the last step in the process” The label in bold reads, “Process stages”. The labels on the timeline (from LTR: Enter telephone number; enter the date of birth; enter the ID number). LTR interpretation: click on the right circle. RTL interpretation: click on the left circle.
Group “Motion”, sub-group “Action performing icons”			
Undo & Redo buttons (I)			“What happens by clicking on the left arrow?”. The label in bold reads, “New form”. LTR interpretation: “Undo the last action”. RTL interpretation: “Redo the action”.
Group “Navigation,” sub-group “Navigation buttons”			
Backward & forward buttons (I)			What do you think will happen after clicking the button “Upload image”?”. The label in bold reads, “Process stages”, the left button label: “< Filling data”; the right button label: “Upload image >”. LTR interpretation: “Move to the next stage”. RTL interpretation: “Back to the previous stage”.
Next & previous buttons (I)			“Which button should be clicked to return to the previous step in the form filling?”. The label in bold reads, “Entering the address”. LTR interpretation: “Left arrow (<)”. RTL interpretation: “Right arrow (>)”.

Legend: (I) Informational task type. (A). Actionable task type *UI item group is marked in bold.

A2 EXAMPLE OF TASK FOR INFORMATION UNDERSTANDING

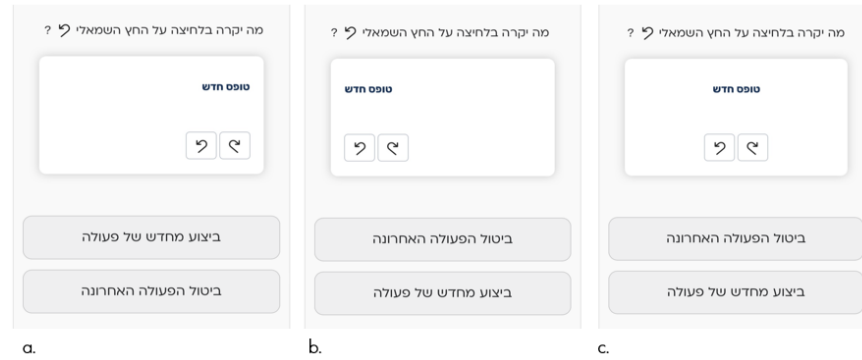


Figure A1: An example of ambiguous stimuli for “Understanding” type tasks. The top text in Hebrew reads “What happens after clicking on the left arrow?”. The label in bold reads “New form.” The answer options: “Undo the last action” and “Redo the action.” The items represent the following conditions (see Section 2.2): (a) Full right, (b) Full left, (c) Center.

A3 EXAMPLE OF TASK FOR ACTION PERFORMING



Figure A2: An example of ambiguous stimuli for “Performance” type tasks. The instruction at the top in Hebrew reads “Enter January 2023.” The label in bold reads “Valid through.” The items represent the following experimental conditions (see Section 2.2): (a) Full right, (b) Full left, (c) Center.

A4 TASK STRUCTURE



Figure A3: (a) Task explanation; (b) UI item (“Date range string” on the print screen); (c) Answer options; (d) “Next” button.

A5 ADDITIONAL EXPLANATION ABOUT EXPERIMENTAL CONDITIONS

Table 3: Experimental conditions

#	Condition	Explanation	Example of stimulus
1	Full right	All UI items appear in their RTL direction version with RTL content alignment.	App 2, fig.1a, App 3, fig.2a
2	Full left	All UI items appear in their LTR direction version with LTR content alignment	App 2, fig.1b, App 3, fig.2b
3	Center	Centered UI item label and context, random RTL or LTR UI item direction versions	App 2, fig.1c, App 3, fig.2c
4	Random	Random representation of stimuli described in the “Full right” and “Full left” conditions	App 2, fig.1a/b, App 3, fig.2a/b

A6 DEMOGRAPHIC DATA

Table 4: Summary about participants' Demographic data

#	Parameter	Value	Arabic sample (%)	Hebrew sample (%)	Total (%)
1	Age	18-24	21.3	14.9	17.3
		25-29	20.5	8.6	12.9
		30-49	47.6	37.3	41.1
		50-64	9.1	24.1	18.7
		65+	1.6	14.7	10.0
2	Gender	Female	65.7	44.9	52.7
		Male	34.3	54.1	47.2
		Other	0	0.1	0.1
3	Native language direction	BiDi	99.4	91.4	94.3
		LTR	0.6	8.6	5.7
4	LTR level proficiency	No	27.3	8.7	15.4
		Starter level	6.8	6.9	6.9
		Excellent in speaking	4.2	4.7	4.5
		Excellent in reading and speaking	15.0	16.3	15.8
		Excellent in writing, reading and speaking	37.8	44.1	41.8
		Native speaker	8.9	19.3	15.5
5	Occupation	Army and forces	1.5	4.5	3.4
		Design	0.8	1.6	1.3
		Editing, translation, copywriting	0	1.5	.9
		Education and research	17.8	10.5	13.1
		Finance, economics	2.4	7.5	5.6
		Government and law	2.9	4.1	3.6
		Health and medicine	10.2	6.3	7.7
		IT	6.3	20.1	15.1
		Maintenance and building	3.1	3.9	3.6
		Management, business or administration	5.8	7.8	7.0
		Not working, pensioner, student	28.8	21.0	23.8
		Service and Sales	4.2	3.9	4.0
		Social, sport & culture	3.2	3.4	3.3
		Other	13.1	4.0	7.3
6	Dominant hand	Left	6.6	9.6	8.6
		Right	93.4	90.3	91.4
7	Preferred UI content language direction	BiDi	83.9	77.1	79.6
		LTR	16.1	22.9	20.4
8	Digital literacy	1	0	0.2	0.1
		2	0.5	0.6	0.55
		3	3.7	1.7	2.7
		4	10.8	3.8	7.3
		5	25.3	13.0	19.6
		6	30.8	32.4	31.6
		7	28.9	48.4	38.7
9	Device	Mobile	85.8	82.2	83.6
		Desktop	14.2	17.5	16.4

A7 TASK FOR “EVENT FLOW IMAGE” UI ITEM, DEMONSTRATED DIFFERENCE IN DIRECTION INTERPRETATION BETWEEN ARABIC AND HEBREW SAMPLES

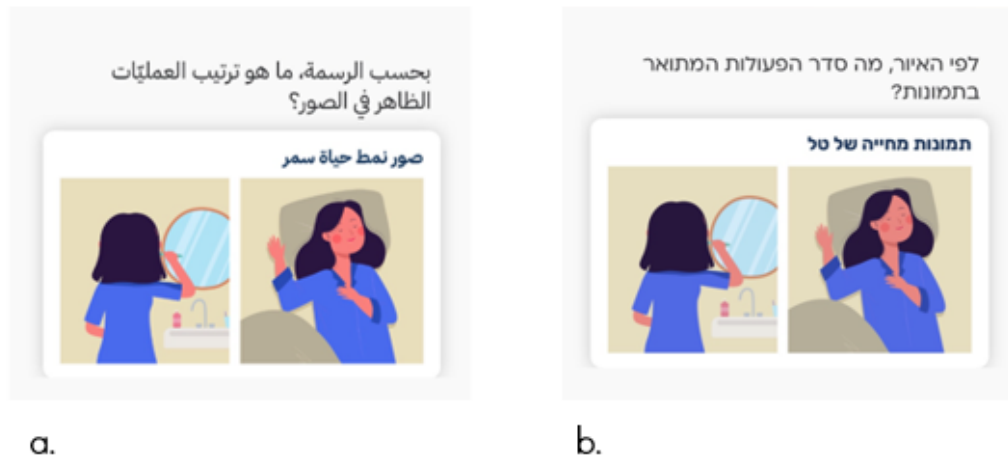


Figure A4: Print screen of the task description and UI item “Event flow image” in “Full right” condition. The explanation text at the bottom in Arabic (a) or Hebrew (b) reads “Based on the illustration, what is the order of actions described by the images?” The label in bold reads “Images from Tal’s life.”